Bhanu Sai Prakash Chikkavarapu

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About Me

Mechanical Engineer and Certified Fluid Power Hydraulic Specialist with 5+ years of experience in hydraulic system design. Skilled in manifold design, component sizing, pumps, cylinders, filtration systems, pressure-drop analysis, and system troubleshooting. Proficient in SolidWorks, Automation Studio, Fixtures/Test Setup and ISO hydraulic schematics.

Professional Experience

Mechanical Design Engineer – Design Cube Engineering Solutions Pvt. Ltd. Jul 2021 – Aug 2024

- Designed hydraulic reservoirs with integrated cooling and filtration systems for mobile fluid power applications.
- Developed hydraulic manifolds and schematics (ISO symbols), reducing hoses and leak paths.
- Sized components such as pumps, motors, heat exchangers, valves based on system requirements.
- Designed industrial electrohydraulic systems, integrating solenoid valves, sensors, and system controls.
- Applied DFMA to improve manufacturability, created CAD models, technical drawings with BOMs.

Graduate Research Assistant – Northern Illinois University Aug 2024 – Present

- Designed a compact and modular fluid-flow test setup, integrating flow control, pressure-drop components.
- Developed fluid-flow schematics and redesigned routing to minimize restriction, leakages and improve system response.
- Designed and evaluated staged filtration assemblies, analyzing flow characteristics, pressure drop, and efficiency.
- Conducted experimental testing and validation to support ongoing R&D and iterative product development.

Design Engineering Intern – Hydraulic Technologies May 2026 – Aug 2026

- Developed a two-stage hi-low high-pressure radial piston pump, optimizing valve operation and internal flow paths.
- Performed competitor product teardown analysis to compare hydraulic architecture and component life ratings.
- Performed burst and buckling studies on high-tonnage aluminum hydraulic cylinders to evaluate failure behavior.
- Designed and fabricated cycle and side-load test fixtures for multiple high-pressure cylinder configurations.

Mechanical Engineer Intern – APSCO, Inc. Jun 2025 – Aug 2025

- Designed hydraulic power units (HPUs) with integrated manifold, D03 directional valves, and filtration systems.
- Developed and fabricated a compact multi-stage inlet air filtration system for mobile blower applications.
- Created custom fixtures to improve assembly efficiency, accelerate prototyping, and ensure repeatable testing.
- Prepared technical manuals covering system operation, installation procedures, and troubleshooting guidelines.

Skills

Fluid Power	**Design & Analysis**	**Manufacturing**	**Testing**	**Software**
Electrohydraulics	Hydraulic Schematics	CNC Machining	Pressure Capacity	SolidWorks
Cartridge valves	Pressure/Flow Analysis	Sheet Metal	Leak/Performance	PLC
Pumps/Motors	FEA Analysis	Additive Manufacturing	System Troubleshooting	MATLAB

Education

Northern Illinois University, IL
Master of Science in Mechanical Engineering 2024 – 2026

VR Siddhartha Engineering College, India
Bachelor of Technology in Mechanical Engineering 2017 – 2021